Extraction	45	0.10
Check up	15	0.20

Simulate the dentist,s clinic for four hours and determine the average waiting time for the patients as well as the idleness of the doctor. Assume that all the patients show up at the clinic at exactly their scheduled arrival time. Starting at 8 A.M. use the following random numbers for handling the above problem:

40, 82, 11, 34, 25, 66, 17 and 79. 12

(c) The output of a production line is checked by an inspector for one or more of three different types of defects, called defects A, B and C. If defect A occurs, the item is scrapped. If defect B or C occurs, the item must be reworked. The time required to rework a B defect is 15 minutes and the time required to rework a C defect is 30 minutes. The probabilities of A, B and C defects are 0.15, 0.20 and 0.10 respectively. For ten items coming off the assembly line, determine the number of items without any defects, the number scrapped and the total minutes of rework time. 12

Use the following random numbers:

RN for defect A : 48 55 91 40 93 01 83 63 47 52

RN for defect B : 47 36 57 04 79 55 10 13 57 09

RN for defect C : 82 95 18 96 20 84 56 11 52 03

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MBA (4th Semester) Examination, April - May, 2016

(New Scheme)

Econometrics and Decision Science

Time Allowed : 3 hours *Maximum Marks :* 80

Minimum Pass Marks : 32

Note : Question *(a)* is compulsory for all the units and carrries 4 marks each.

Solve any one question from (*b*) and (*c*) from each unit which carries 12 marks each.

Unit-I

1. (*a*) Explain the Methodology of Econometrics. 4

(*b*) Given below is a transaction matrix showing an input-output relationship :

(million rupees)

Producing sector	Consuming sector			Final Demand	Total output
	Steel	Transport	Power		
Steel	20	80	60	240	400
Transport	60	40	200	500	800
Power	100	200	40	60	400
Primary Input (Labour)	220	480	100		800

What will be the change in output when final demand of steel sector grows from 240 to 250? 12

(*c*) What is input-output analysis? Write a note on the assumptions of input-output analysis. 12

Unit-II

2. (*a*) What are major limitations of Game Theory? 4

(*b*) Solve the Game : 12

$$\begin{array}{cc} & \text{Player B} \\ & \begin{array}{ccc} b_1 & b_2 & b_3 \end{array} \\ \text{Player A} \rightarrow \begin{array}{c} a_1 \\ a_2 \\ a_3 \end{array} & \begin{bmatrix} 7 & 1 & 7 \\ 9 & -1 & 1 \\ 5 & 7 & 6 \end{bmatrix} \end{array}$$

(*c*) In an election of M.L.A, two political parties A and B are thinking of nominating a candidate in a closed session. Whose results are to be announced simultaneously. The following odds are offered for the various possible combinations of candidates : 12

Party A	Odds	Party B
Sharma	3:1	Singh
Sharma	4:1	Gill
Sharma	1:3	Bajwa
Goel	3:7	Singh
Goel	3:2	Gill
Goel	1:4	Bajwa
Kapoor	4:1	Singh
Kapoor	1:4	Gill
Kapoor	1:3	Bajwa

The parties want to select candidate in accordance with standard minimax criterion. What are the optimal strategies for party A and B?

Unit-III

3. (a) Is the assumption of stationary transition probabilities realistic, in your opinion? Why or why not? 4

(b) A student tries to be punctual for the classes. If he is late on a day, he is 90 percent sure to be on time next day. Similarly, if he is on time, then there is 30 percent chance that he will be late on the next day. How often, in the long run, is he expected to be late for the class? 12

(c) A manufacturing company has a certain piece of equipment that is inspected at the end of each day and classified as just overhauled, good, fair or inoperative. If the item is inoperative it is overhauled, a procedure that takes one day. Let us denote the four classifications as states 1,2,3 and 4 respectively. Assume that the working condition of the equipment follows as Markov chain with the following transition matrix : 12

$$P = \text{Today} \begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \end{array} \begin{array}{c} \text{Tomorrow} \\ \begin{array}{cccc} 1 & 2 & 3 & 4 \end{array} \\ \begin{bmatrix} 0 & 3/4 & 1/4 & 0 \\ 0 & 1/2 & 1/2 & 0 \\ 0 & 0 & 1/2 & 1/2 \\ 1 & 0 & 0 & 0 \end{bmatrix} \end{array}$$

If it costs Rs. 125 to overhaul a machine (including lost time) on the average and Rs. 75 as production lost if a machine is found inoperative. Using the steady-state probabilities, compute the expected per day cost of maintenance. 12

Unit-IV

4. (*a*) How are maximum likelihood and expectation principles of choice differentiated? 4

(*b*) A certain output is manufactured at Rs. 8 and sold at Rs. 14 per unit. The product is such that if it is produced but not sold during a daytime it becomes worthless. The daily sales records in the past are as follows :

Demands per day :	30	40	50	60	70
Sales level was recorded :	24	24	36	24	12

(*i*) Calculate the average expected sales of a day.

(*ii*) Also find the value of perfect information. 12

(*c*) There is 40% chance that a patient admitted to the hospital is suffering from cancer. A doctor has to decide whether a serious operation should be performed or not. If the patient is suffering from cancer and the serious operation is performed, the chance that he will recover is 70%, otherwise it is 35% on the other hand, if the patient is not suffering from cancer and the serious operation is performed, the chance that he will recover is 80%, otherwise it is 100%. Assume that recovery and death are the only possible results.

Construct an appropriate decision tree. What decision should the doctor take? 12

Unit-V

5. (*a*) Write a note on Monte Carlo Simulation? 4

(*b*) A dentist schedules all her patients for 30 min. appointments. Some of the patients take more or less than 30 minutes depending on the type of dental work to be done. The following summary shows the various categories of work, their probabilities and the time needed to complete the work :

Category	Time required (minutes)	Probability of Category
Filling	45	0.40
Crown	60	0.15
Cleaning	15	0.15